Homework 3: Project update
Due: 1 December 2025

Assignment

1. Please revise the abstract to your project. Answer the following questions.
   (a) What is the problem you are solving?
   (b) Why or to whom is it important or interesting?
   (c) What is your strategy?
   (d) What outcome to you expect?

   In part (d), you can write about an idealized outcome, even if you have not yet achieved it. It's okay if this changes later.

2. Flesh out out the answers to each of the questions (a)-(d) with 1-3 paragraphs. Feel free to include details about datasets, models, and results in progress. When answering these, you may use the TA's tips below.

3. Summarize your plans for the final weeks, and what bottlenecks you are facing.

TA's tips

Here are some tips from the TAs in thinking about your project's progress report (and thinking about your project in general).

Your project will involve a probabilistic model (or class of models). For a clear description of your project, state the model and its joint probability. You can also illustrate the the graphical model with plate notation.

If your project applies probabilistic modeling to a real-world problem, define the model in terms of the relevant observed and latent variables in that context.

Here are some details to consider when describing your project.

- For **real-world problems:**
  - **Data:** What dataset do you plan to use, where it comes from, the number of observations, and the key variables it contains. Does it require preprocessing steps? (conceptually, not at the level of coding details). How will you split the data for training, validation, and testing? How does this setup align with your project goals?
  - **Estimation and inference:** What are you trying to approximate? What algorithms will you use?
  - **Evaluation:** How will you assess your project? Think about clear, measurable criteria for success. Describe the metrics, benchmarks, or comparisons you will use for evaluation.
- For **algorithmic projects:**
  - **Setting:** Clearly describe the probabilistic modeling setting in which your algorithm operates. What problem are you trying to solve?
  - **Algorithm:** Present the algorithm you plan to develop. Provide a high-level description of the method, the key idea behind it, and how it differs from existing approaches.
  - **Evaluation:** Describe how you will assess the algorithm's performance. Specify evaluation metrics and benchmarks (e.g., accuracy of posterior approximation, speed, scalability, robustness), and explain which baselines you will compare to. If you plan synthetic experiments, describe the data-generation setup; if you will also test on real data, state which datasets and why.
- For **theoretical projects:**
  - **Setting:** Clearly state the mathematical problem you are studying and the class of probabilistic models it concerns. Formalize the assumptions under which you are working.

  - **Objective:** Specify the theoretical question you aim to answer. Examples include characterizing identifiability, deriving consistency or convergence rates, establishing optimality properties, or analyzing model expressiveness.
  - **Approach:** Outline the mathematical tools and techniques you plan to use (e.g., concentration inequalities, variational analysis, coupling arguments). If applicable, describe simpler toy settings or special cases that you will first analyze to build intuition.
  - **Expected results:** State the main results you expect or aim to prove. Formulate them in a clear manner, emphasizing their meaning and implications for probabilistic modeling or inference.
- For **other projects:** If your project does not fall into any of these categories above, clearly summarize the problem you are addressing, the approach you plan to take, and how you will evaluate your progress and success. Be explicit about your goals and the criteria you will use to assess whether they are met.